

ACT-542: Applied Regression

Multiple Linear Regression Analysis

Review of Previous Lecture

In last lecture we discussed:

- Simple Linear Regression Model
- Method of Ordinary Least Squares (OLS)
- Properties of OLS
- Standard Error of Regression (SER)
- Coefficient of Determination (R^2)
- An Example

Objectives of Current Lecture

In the current lecture:

- An example of Simple Linear Regression Model
- Introduction Multiple Linear Regression Model
- Assumptions of Simple & Multiple Regression Models
- Estimation and Interpretation

Regression: An Example

Example: Given the data on X and Y.

- Compute Least squares regression equation of Y on X.
- Interpret regression coefficients.
- Show that $\sum_t (y_t - \hat{y}_t) = 0$
- Compute SER.
- Calculate Coefficient of Determination (R^2).

X	Y
5	16
6	19
8	23
10	28
12	36
13	41
15	44
16	45
17	50

Solution:

Do demo in Excel.

Multiple Linear Regression Model

- General form of a Multiple Linear Regression Model:

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \dots + \beta_k x_{kt} + u_t \quad t=1,2,\dots,n$$

Where,

y_i 's are observations on dependent variable (y)

x_i 's ($i=1,2,\dots,k$) are k regressors each having n observations

β_i 's ($i=1,2,\dots,k$) are regression coefficients or parameters

u_i 's are errors

Multiple Linear Regression Model

Assumptions of Linear Regression Model

- Mean of error term is zero, i.e. $E(u_t)=0$, $t=1,2,\dots,n$
- Homoskedasticity: Variance of error term is constant,

$$\text{Var}(u_t)=\sigma^2, t=1,2,\dots,n$$

- No Serial or Autocorrelation: Error terms are independent of each other,

$$E(u_i, u_j)=0, \text{ for all } i \neq j$$

Multiple Linear Regression Model

Assumptions of Linear Regression Model

- Regressor (X) and error term (u) are independent of each other, i.e.
 $E(X, u_i) = 0$
- Error are normally distributed with a mean of zero and a constant variance, i.e.

$$u_t \sim N(0, \sigma^2)$$

- No Multicollinearity between regressor

Multiple Linear Regression Model

Estimating Least Squares Estimates via Normal Equation

Simple linear regression model: $Y = \alpha + \beta X + U$

Estimated regression model is: $Y = \alpha + \beta X$

Normal Equation are:

$$\sum Y = n\alpha + \beta \sum X$$

$$\sum XY = \alpha \sum X + \beta \sum X^2$$

Multiple Linear Regression Model

Normal Equation are:

$$\sum Y = n\alpha + \beta \sum X$$

$$\sum XY = \alpha \sum X + \beta \sum X^2$$

Solving normal equations simultaneously, we get:

$$\beta = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\alpha = \frac{\sum x^2 \sum y - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$$

So estimated regression line is:

$$Y = \alpha + \beta X$$

Multiple Linear Regression Model

Estimating Least Squares Estimates via Normal Equation

Multiple linear regression model: $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + U$

Estimated regression model is: $Y = \alpha + \beta_1 X_1 + \beta_2 X_2$

Normal Equation are:

Multiple Linear Regression Model

Normal Equation are:

$$\sum Y = n\alpha + \beta_1 \sum X_1 + \beta_2 \sum X_2$$

$$\sum X_1 Y = \alpha \sum X_1 + \beta_1 \sum X_1^2 + \beta_2 \sum X_1 X_2$$

$$\sum X_2 Y = \alpha \sum X_2 + \beta_1 \sum X_1 X_2 + \beta_2 \sum X_2^2$$

Multiple Linear Regression Model

Normal Equation are:

$$\sum Y = n\alpha + \beta_1 \sum X_1 + \beta_2 \sum X_2$$

$$\sum X_1 Y = \alpha \sum X_1 + \beta_1 \sum X_1^2 + \beta_2 \sum X_1 X_2$$

$$\sum X_2 Y = \alpha \sum X_2 + \beta_1 \sum X_1 X_2 + \beta_2 \sum X_2^2$$

Solving normal equations simultaneously, we get:

$$\alpha = \bar{Y} - \beta_1 \bar{X}_1 - \beta_2 \bar{X}_2$$

$$\beta_1 = \frac{(\sum X_1 Y)(\sum X_2^2) - (\sum X_2 Y)(\sum X_1 X_2)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2}$$

$$\beta_2 = \frac{(\sum X_2 Y)(\sum X_1^2) - (\sum X_1 Y)(\sum X_1 X_2)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2}$$

Multiple Linear Regression Model

Example:

A statistician wants to predict the incomes of restaurants, using two independent variables, the number of restaurant employees and restaurant floor area. He collected the following data:

Calculate:

Estimated linear regression equation.

Y	X ₁	X ₂
30	10	15
22	5	8
16	10	12
7	3	7
14	2	10

Solution:

Do the demo in Excel